

09/04/21

HW 1) Find out the application of recursion.

Is recursion still used in the present programming if so where?

2) What are the programming languages that support recursive calls.

1) Recursion is still widely used in present programming recursion means, a function calls itself to perform the given task.

Application :-

- ① Tree data structure [Binary Trees, AVL trees]
- ② graph algorithms [DFS, Topological sorting]
- ③ divide and conquer [merge sort, quick sort, binary search].
- ④ Backtracking problems - [sudoku solving]

- ⑤ File system operations
- ⑥ Artificial intelligence
- ⑦ Dynamic programming [Fibonacci]
- ⑧ Mathematical computations [Factorial, Tower of Hanoi]

2) Yes, Recursion is used in

- Artificial intelligence (AI)
- game developing
- operating system (OS)
- compiler
- data structures (DSA)
- Machine learning algorithms

3) what are the programming languages that support recursive calls.

Ans. C, C++, Java, Python, JavaScript, C++,

Ruby, PHP.